

27/08/24

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
Department of Electrical Engineering
MINI TEST(August 2024)

Sub: EMF & Materials (E-211)

Time: 1 hr.

Max. Marks: 20

Q. No.	Question	Marks
1	Transform each of the following vectors to spherical coordinates at the point specified (a) $5\hat{x}$ at point $(r = 4, \theta = 25^\circ, \phi = 120^\circ)$ (b) $5\hat{r}$ at point $(x=2, y=3, z=-1)$, all linear coordinates are in meter.	06
2	Find electric field at $P(1, 1, 1)$ caused by four identical 3nC point charges located at $P_1(1, 1, 0)$, $P_2(-1, 1, 0)$, $P_3(-1, -1, 0)$ and $P_4(1, -1, 0)$. All the coordinates are in rectangular system.	09
3	Compute the divergence of the vector function $F = \rho(2 + \sin^2\phi)\hat{\rho} + (\rho\sin\phi\cos\phi)\hat{\phi} + 3z\hat{z}$	05

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